Chapter 2
Discrete Time Markov Chain: Part I

- Defintion
- Chapman-Kolmogorov equations
- Classification of states
- Long-run proportions

Defintion

- Consider a discrete-time stochastic process $\{X_n : n = 0, 1, 2, ..., \}$, taking values from $\{0, 1, 2, ...\}$.
- It is nature that the distribution of X_{n+1} should depend on the history X_n, X_{n-1},...,X_0.
- If the distribution of X_{n+1} depends *only* on X_n for all $n \geq 0$, we say the process X_n is a Markov chain. In detail,

  $$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, ..., X_0 = i_0) = P(X_{n+1} = j | X_n = i) = P_{ij}.$$

- The value P_{ij} is called the one-step transition probability from state i to state j. Obviously,

  $$P_{ij} \geq 0, \sum_{j=0}^{\infty} P_{ij} = 1, \text{ for all } i, j \geq 0.$$

  The matrix $\mathbf{P} = (P_{ij})$ is called the transition probability matrix of X_n.

Prop. Let $1 \le k \le n_0 < n_1 < n_2 < \cdots < n_m$. Then the conditional dist. of $X_{n_1}, X_{n_2}, \cdots, X_{n_m}$ given X_{n_0} is the same as $X_{n_1-k}, X_{n_2-k}, \cdots, X_{n_m-k}$ given X_{n_0-k}.

Proof. Write

$$P(X_{n_1} = k_{n_1}, X_{n_2} = k_{n_2}, \cdots, X_{n_m} = k_m \mid X_{n_0} = k_{n_0})$$

$$= \sum_{\substack{k_i \ge 0 \\ n_0+1 \le i \le n_m \\ i \neq n_1, n_2, \cdots, n_m}} P(X_i = k_i,\ n_0+1 \le i \le n_m \mid X_{n_0} = k_{n_0}) \quad (*)$$

Easily, we have: let $A_0, A_1, \cdots, A_m$ be events, then

$$P(A_m A_{m-1} \cdots A_1 \mid A_0) = P(A_m \mid A_0 \cdots A_{m-1}) \cdot P(A_{m-1} \mid A_0 \cdots A_{m-2}) \cdots P(A_1 \mid A_0)$$

Then the probab in $(*)$ is equal to

$$P(k_{n_0}, k_{n_1}) \cdot P(k_{n_1}, k_{n_2}) \cdots P(k_{n_{m-1}}, k_{n_m}).$$

Suming this over the indices k_i as in $(*)$ has nothing to do with $n_1, \cdots, n_m$, but depends on the distances $n_1 - n_0,\ n_2 - n_1,\ n_3 - n_2, \cdots, n_m - n_{m-1}$.

- $P(X_3 = b \mid X_1 = a)$

$$= \sum_c P(X_3 = b, X_2 = c \mid X_1 = a)$$

$$= \sum_c P(a, c) P(c, b)$$

This is because

$$P(X_3 = b, X_2 = c \mid X_1 = a)$$

$$= P(X_3 = b \mid X_2 = c, X_1 = a)\, P(X_2 = c \mid X_1 = a)$$

$$= P(X_3 = b \mid X_2 = c) \cdot P(X_2 = c \mid X_1 = a)$$

- $P(X_7 = b \mid X_5 = a)$

$$= \sum_c P(X_7 = b, X_6 = c \mid X_5 = a)$$

$$= \sum_c P(a, c) P(c, b)$$

Thus $P(X_3 = b \mid X_1 = a) = P(X_{3+\ell} = b \mid X_{1+\ell} = a)$ for any $\ell \geq 0$.

Examples

- **Example 4.3** On any given day Gary is either cheerful (C), so-so (S), or glum (G). If he is cheerful today, then he will be C, S, or G tomorrow with respective probabilities 0.5, 0.4, 0.1. If he is feeling so-so today, then he will be C, S, or G tomorrow with probabilities 0.3, 0.4, 0.3. If he is glum today, then he will be C, S, or G tomorrow with probabilities 0.2, 0.3, 0.5.

State 0 = C
State 1 = S
State 2 = G

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} \\ P_{10} & P_{11} & P_{12} \\ P_{20} & P_{21} & P_{22} \end{bmatrix} = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$$

Examples

- **Example 4.5 (A Random Walk Model)** A Markov chain whose state space is given by the integers $i = 0, \pm 1, \pm 2, ...$ is said to be a random walk if, for some number $0 < p < 1$,

$$P_{i,i+1} = p = 1 - P_{i,i-1}, i = 0, \pm 1, ...$$

Examples

- **Example 4.6 (A Gambling Model)** Consider a gambler who, at each play of the game, either wins \$1 with probability p or loses \$1 with probability $1-p$. If we suppose that our gambler quits playing either when he goes broke or he attains a fortune of N, then the gambler's fortune is a Markov chain having transition probabilities

$$P_{i,i+1} = p = 1 - P_{i,i-1}, i = 1, 2, ..., N-1, \text{and } P_{00} = P_{NN} = 1.$$

States 0 and N are called absorbing states

$q = 1 - p$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ q & 0 & p & 0 & 0 & \cdots & 0 \\ 0 & q & 0 & p & 0 & \cdots & 0 \\ 0 & 0 & q & 0 & p & \cdots & 0 \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ 0 & & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

This is a $(N+1) \times (N+1)$ matrix

Transforming a Process into a Markov Chain

- **Example 4.4** Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days, then it will rain tomorrow with probability 0.2.

- If we let the state at time n depend only on whether or not it is raining at time n, then the preceding model is not a Markov chain (why not?).
- However, we can transform this model into a Markov chain by saying that the state at any time is determined by the weather conditions during both that day and the previous day.

Examples

- **Example 4.7** In most of Europe and Asia annual automobile insurance premiums are determined by use of a *Bonus Malus* (Latin for Good-Bad) system. Each policyholder is given a positive integer valued state and the annual premium is a function of this state. A policyholder's state changes from year to year in response to the number of claims made by that policyholder.
- Suppose the number of claims made by a particular policyholder is Poisson with mean λ.

State	**Annual Premium**	**Next state if**			
		0 claims	***1 claim***	***2 claims***	***$\geqslant$ 3 claims***
1	200	1	2	3	4
2	250	1	3	4	4
3	400	2	4	4	4
4	600	3	4	4	4

- Find a Markov chain model for the policyholder's state over years.

$$\begin{cases} p_{11} = P(0 \text{ claims}) = e^{-\lambda} \\ p_{12} = P(1 \text{ claim}) = \lambda e^{-\lambda} \\ p_{13} = P(2 \text{ claim}) = \frac{1}{2}\lambda^2 e^{-\lambda} \\ p_{14} = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{1}{2}\lambda^2 e^{-\lambda} \end{cases}$$

$$X \sim Poi(\lambda)$$

$$a_k = P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

$$k = 0, 1, 2, 3, \dots$$

$$\begin{cases} p_{21} = P(0 \text{ claim}) = e^{-\lambda} \\ p_{22} = 0 \\ p_{23} = P(1 \text{ claim}) = \lambda e^{-\lambda} \\ p_{24} = P(X \geq 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} \end{cases}$$

$$\vdots$$

$$\begin{cases} p_{41} = 0 \\ p_{42} = 0 \\ p_{43} = e^{-\lambda} \\ p_{44} = 1 - e^{-\lambda} \end{cases}$$

$$P = \begin{pmatrix} a_0 & a_1 & a_2 & 1 - a_0 - a_1 - a_2 \\ a_0 & 0 & a_1 & 1 - a_0 - a_1 \\ 0 & a_0 & 0 & 1 - a_0 \\ 0 & 0 & a_0 & 1 - a_0 \end{pmatrix}$$

Chapman-Kolmogorov Equations

- Now we consider how to compute n-step transition probabilities of a Markov chain. Define
$$P_{ij}^n = P(X_n = j | X_0 = i).$$
- Starting from the simple $n = 2$ case, we have
$$P_{ij}^2 = \sum_k P_{ik} P_{kj} = \mathbf{P}^{(2)}(i, j).$$
- In general, we have
$$P_{ij}^{m+n} = \sum_k P_{ik}^m P_{kj}^n, \text{ for all } m, n \geq 1.$$
These equations are called Chapman-Kolmogorov equations. By these equations, we can prove that
$$P_{ij}^n = \mathbf{P}^{(n)}(i, j), \text{ for all } n \geq 1.$$

$$P_{ij}^{m+n} = P(X_{m+n} = j \mid X_0 = i)$$

$$= \sum_{k=0}^{\infty} P(X_{m+n} = j, X_n = k \mid X_0 = i)$$

$$= \sum_{k=0}^{\infty} P(X_{m+n} = j \mid X_n = k, X_0 = i) P(X_n = k \mid X_0 = i)$$

$$= \sum_{k=0}^{\infty} P_{ik}^{n} P_{kj}^{m}$$

$$A = (a_{ij})_{p \times q}, \quad B = (b_{ij})_{q \times r}$$

Let $C = (c_{ij})_{p \times r} = AB$. Then

$$C_{ij} = \sum_{k=1}^{q} a_{ik} b_{kj}$$

$\therefore$ Let $P^{(n)} = (P_{ij}^{n})$. obviously,

$$P^{(m+n)} = P^{(m)} \cdot P^{(n)}$$

$\therefore$ $P^{(2)} = P \cdot P$ $\quad P^{(3)} = P^{(2)} \cdot P^{(1)} = P^3$

$$P^{(n)} = P^n$$

Prop. The conditional dist. of $X_{n_1}, X_{n_2}, \cdots, X_{n_m}$ given X_{n_0}, where $1 \le k \le n_0 < n_1 < n_2 < \cdots < n_m$, is the same as $X_{n_1-k}, X_{n_2-k}, \cdots, X_{n_m-k}$ given X_{n_0-k}.

Proof. Write

$$P(X_{n_1} = k_{n_1}, X_{n_2} = k_{n_2}, \cdots, X_{n_m} = k_m \mid X_{n_0} = k_{n_0})$$

$$= \sum_{\substack{k_i \ge 0 \\ n_0+1 \le i \le n_m \\ i \ne n_1, n_2, \cdots, n_m}} P(X_i = k_i,\ n_0+1 \le i \le n_m \mid X_{n_0} = k_{n_0}) \quad (*)$$

Easily, we have: let $A_0, A_1, \cdots, A_m$ be events, then

$$P(A_m A_{m-1} \cdots A_1 \mid A_0) = P(A_m \mid A_0 \cdots A_{m-1}) \cdot P(A_{m-1} \mid A_0 \cdots A_{m-2}) \cdots P(A_1 \mid A_0)$$

Then the probab in $(*)$ is equal to

$$P(k_{n_0}, k_{n_1}) \cdot P(k_{n_1}, k_{n_2}) \cdots P(k_{n_{m-1}}, k_{n_m}).$$

Suming this over the indices k_i as in $(*)$ has nothing to do with $n_1, \cdots, n_m$, but depends on the distances of $n_1 - n_0,\ n_2 - n_1,\ n_3 - n_2, \cdots, n_m - n_{m-1}$.

Examples

- **Example 4.8** Suppose that the chance of rain tomorrow depends on previous weather conditions only through whether or not it is raining today and not on past weather conditions. Suppose also that if it rains today, then it will rain tomorrow with probability 0.7, and if it does not rain today, it will rain tomorrow with probability 0.4. Find the probability that it will rain four days from today given that it is raining today.

0.7

$$P = \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \end{array}$$

0 = rain

1 = no rain

$p_{00}^4 = ?$

$$P^2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix}.$$

$$P^4 = \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix} \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix} = \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{bmatrix} \end{array}.$$

Answer = 0.5749

Examples



- **Example 4.10** An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces. If initially both balls are red, find the probability that the fifth ball selected is red.

X_n = # of red balls in the urn after nth action

$X_n = 0, 1, 2$

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{bmatrix} \end{array}$$

$\bigcirc\bigcirc \rightarrow OO : \frac{1}{2} \times 0.2 = 0.1$

p(fifth selection is red) is NOT replaced

$$= \sum_{i=0}^{2} p(\text{fifth selection is red} \mid X_4 = i)\, \underbrace{p(X_4 = i)}_{p(X_4 = i \mid X_0 = 2)}$$

$$= 0 \cdot P_{20}^4 + 0.5 \cdot P_{21}^4 + 1 \cdot P_{22}^4$$

Calculate $P^{(4)}$: $P_{21}^4 = 0.4352$, $P_{2,2}^4 = 0.4872$

$\therefore$ p(The fifth selection is red) $= 0.7048$.

$$P^4 = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0.4872 & 0.4352 & 0.0776 \\ 0.2176 & 0.5648 & 0.2176 \\ 0.0776 & 0.4352 & 0.4872 \end{bmatrix} \end{array}$$

Absorbing Probability

- Consider a Markov chain with transition probabilities P_{ij}. Let $\mathcal{A}$ be a set of states, and suppose we are interested in the probability that the Markov chain ever enters any of the states in $\mathcal{A}$ by time m.That is,for a given state $i \notin \mathcal{A}$,we are interested in determining

$$\beta = P(X_k \in \mathcal{A} \text{ for some } k = 1, ..., m | X_0 = i)$$

- To determine the preceding probability we will define a Markov chain $\{W_n : n \geq 0\}$ whose states are the states that are not in $\mathcal{A}$ plus an additional state, which we will call $\mathcal{A}$. Once the W_n enters state $\mathcal{A}$ it remains there forever.

Absorbing Probability

- **Example 4.13** Consider a Markov chain with states 1, 2, 3, 4, 5, and suppose that we want to compute

$$P(X_4 = 2, X_3 \leq 2, X_2 \leq 2, X_1 \leq 2 | X_0 = 1).$$

All we know about the transition probabilities are:

$$P_{11} = 0.3, P_{12} = 0.3, P_{21} = 0.1, P_{22} = 0.2.$$

Probab that starting at 1 and end at 4 in 4th step but never get into {3.4.5} in between

$$P(X_4 = 2, X_3 \leq 2, X_2 \leq 2, X_1 \leq 2 | X_0 = 1).$$

$$= \sum_{1 \leq i,j,k \leq 2} P(X_4 = 2, X_3 = k, X_2 = j, X_1 = i \mid X_0 = 1)$$

$$= \sum_{1 \leq i,j,k \leq 2} p_{1i} p_{ij} p_{jk} p_{k2}$$

$$= 0.3^3 \cdot 0.3 + 0.3^2 \cdot 0.3 \cdot 0.3 + \cdots$$

1 1 1
1 1 2
1 2 1
1 2 2
2 1 1
2 1 2
2 2 1
2 2 2

Long-run Behavior of Markov Chain

- Consider a Markov chain $\{X_n : n = 0, 1, 2, ...\}$ with no absorbing state.

- We are interested in the following questions:
  - Does the limit $\frac{1}{N}\sum_{n=1}^{N} X_n$ exist?
  - Does the limit $\frac{1}{N}\sum_{n=1}^{N} 1(X_n = i)$ exist?
  - Does the limit $P(X_n = \cdot | X_0 = i)$ exist? Does it depend on the initial state $X_0 = i$?

- To answer the above questions, we need to learn the following notions:
  - Classification of states
  - Recurrence
  - Limiting probability

Classification of States

- State j is said to be accessible from state i if $P_{ij}^n > 0$ for some $n \geq 0$. Note that this implies that state j is accessible from state i if and only if, starting in i, it is possible that the process will ever enter state j.
- Two states i and j that are accessible to each other are said to communicate, and we write $i \leftrightarrow j$.
- **Example 4.16.** Consider a Markov chain consisting of the four states 0, 1, 2, 3 and having transition probability matrix

$$\mathbf{P} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \left\| \begin{array}{cccc} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 1 \end{array} \right\| \end{array}$$

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$$\begin{pmatrix} a & 0 \\ b & c \end{pmatrix}\begin{pmatrix} a & 0 \\ b & c \end{pmatrix} = \begin{pmatrix} a^2 & 0 \\ ab+bc & c^2 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a^2 & ab+bc \\ 0 & c^2 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\begin{pmatrix} \frac{1}{a} & -\frac{b}{ca} \\ 0 & \frac{1}{c} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix};\ a \neq 0,\ c \neq 0 \right\} \text{ is a group}$$

block-matrix

$$P^n = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ a & b & c & d \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}, \quad n \geq 1,$$

where $a, b, c, d > 0$.

friends = {0, 1}

- 0 and 1 are cnmate.
- 0, 1, 3 are all accessible from 2.
- 2 is not accessible from 0, 1, or 3.
- Neither of 0, 1, 2 is accessible from 3.
- 3 is an absorbing state.

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Classification of States

- The relation of communication satisfies the following three properties:
  1. Each state i communicates with itself.
  2. If state i communicates with state j, then state j communicates with state i.
  3. If state i communicates with state j, and state j communicates with state k, then state i communicates with state k.
- Then, we divide the state space into disjoint classes by putting states that communicate with each other into the same class.

- A Markov chain is called irreducible if its state space has only one class and all the states communicate.

communication is an equivalent relation

$p_{ij}^{m} > 0$, $p_{jk}^{n} > 0$. Then

$$p_{ik}^{m+n} = \sum_{r} p_{ir}^{m} p_{rk}^{n}.$$



$\therefore \quad p_{ik}^{m+n} \geq p_{ij}^{m} \cdot p_{jk}^{n} > 0.$

Examples

- Is a random walk irreducible? Yes

- Is the Markov chain in the gambling model irreducible? No

Recurrent and Transient States

- For any state i we let f_i denote the probability that, starting in state i, the process will ever reenter state i in finite steps. State i is said to be recurrent if $f_i = 1$ and transient if $f_i < 1$.
- The Markov property that the distribution of X_{n+1} depends only on X_n leads to the following two facts:
  1. If state i is recurrent then, starting in state i, the process will reenter state i again and again and again — in fact, infinitely often.
  2. If state i is transient then, starting in state i, the number of time periods that the process will be in state i has a *geometric* distribution with finite mean $1/(1 - f_i)$.

Y = # of times X_n visit i Starting in i. $Y = 1, 2, 3, \cdots$

Then $Y \sim Geo(1-f_i)$, i.e.,

$P(Y = k) = (1-f_i) f_i^{k-1}$, $k = 1, 2, \cdots$

$P(X_n = i \text{ for some } n \geq 1 \mid X_0 = i) = f_i$.

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Let T_1 be the 1st time X_n visits i, otherwise $= \infty$

" T_2 " 2nd " " " "

" T_3 " 3rd " " " "

We will show $T_1,\ T_2 - T_1,\ T_3 - T_2, \cdots$ are iid with

$P(T_1 < \infty) = f_1$. Let $Y = \#$ of times the chain visits i. $Y \geq 1$ since $X_0 = i$.

Then $P(Y = 2) = P(T_1 < \infty, T_2 = \infty) = P(T_1 < \infty, T_2 - T_1 = \infty)$

$= P(T_1 < \infty) - P(T_1 < \infty, T_2 - T_1 < \infty)$

$= f_1(1 - f_1)$

$P(Y = m) = P(T_1 < \infty, \cdots T_{m-1} < \infty, T_m = \infty)$

$= P(T_1 < \infty, T_2 - T_1 < \infty, \cdots, T_{m-1} - T_{m-2} < \infty, T_m - T_{m-1} = \infty)$

$= P(T_1 < \infty, T_2 - T_1 < \infty, \cdots, T_{m-1} - T_{m-2} < \infty) -$

$P(T_1 < \infty, T_2 - T_1 < \infty, \cdots, T_m - T_{m-1} < \infty)$

$= f_i^{m-1} - f_i^m = f_i^{m-1}(1 - f_i)$.

$T_2 - T_1$ and T_1 are indept: By induction, we only illustrate the following case.

$P(T_2 - T_1 = m, T_1 = n) = P(T_1 = n) \cdot P(T_2 = m+n \mid T_1 = n)$

$P(T_2 = m+n \mid T_1 = n) = P(X_{n+1} \neq i, \cdots, X_{m+n} = i \mid X_0 = i, X_1 \neq i, \cdots, X_{n-1} \neq i, X_n = i)$

$= P(X_{n+1} \neq i, \cdots, X_{m+n} = i \mid X_n = i)$

$= P(X_1 \neq i, \cdots, X_{m-1} \neq i, X_m = i \mid X_0 = i)$

$= P(T_1 = n)$.

Thus, $P(T_1 = n_1, T_2 - T_1 = n_2, \cdots, T_k - T_{k-1} = n_k) = \prod_{j=1}^{k} P(T_1 = n_j),\ \forall\ n_1, \cdots, n_k$

Recurrent and Transient States

- **Proposition 4.1:** State i is

$$\begin{aligned} &\text{recurrent,} \quad \text{if } \sum_{n=1}^{\infty} P_{ii}^n = \infty; \\ &\text{transient,} \quad \text{if } \sum_{n=1}^{\infty} P_{ii}^n < \infty. \end{aligned}$$

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- **Corollary 4.2:** If state i is recurrent and state j communicates state i, state j is also recurrent.

As a result, all states of a finite irreducible Markov chain is recurrent.

- **Corollary 4.2:** If state i is recurrent and state j communicates state i, state j is also recurrent.

Proof. Since $i \sim j$ and $j \sim i$, there exist $k \geq 1, m \geq 1$ s.t. $p_{ij}^{k} > 0$, $p_{ji}^{m} > 0$. Thus, by the Chapman-Kolmogorov eq,

$$p_{jj}^{m+n+k} = \sum_{s,t} p_{js}^{m} p_{st}^{n} p_{tj}^{k} \geq p_{ji}^{m} p_{ii}^{n} p_{ij}^{k}.$$

$$\therefore \quad \sum_{n=1}^{\infty} p_{jj}^{n} \geq p_{ij}^{k} \sum_{n=1}^{\infty} p_{ii}^{n} \cdot p_{ji}^{m} = \infty.$$

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$$\sum_{n=1}^{\infty} p_{ii}^n = \sum_{n=1}^{\infty} P(X_n = i \mid X_0 = i)$$

$$= E \sum_{i=1}^{\infty} 1(X_n = i)$$

$$= E V_i,$$

$P_i = P(\cdot \mid X_0 = i)$

E_i = expectation w.r.t. P_i

$1(A) = \begin{cases} 1, & w \in A; \\ 0, & w \notin A. \end{cases}$

where V_i = # of times X_n visit i.

Use $EV_i = \sum_{k=1}^{\infty} P(V_i \geq k)$ and

$$P(V_1 \geq 1) = P(T_1 < \infty) = f_i$$

$$P(V_1 \geq k) = P(T_1 < \infty, T_2 < \infty, \cdots, T_k < \infty)$$

$$= P(T_1 < \infty, T_2 - T_1 < \infty, \cdots, T_k - T_{k-1} < \infty)$$

$$= f_i^k$$

by the slide before the previous one. Thus

$$EV_i = \sum_{k=1}^{\infty} f_i^k = \begin{cases} \infty, & \text{if } f_i = 1; \\ \frac{f_i}{1-f_i}, & \text{if } f_i < 1. \end{cases}$$

Thus,

$$\sum_{i=1}^{\infty} p_{ii}^n = \begin{cases} \infty & \text{iff } f_i = 1; \\ \frac{f_i}{1-f_i} & f_i < 1. \end{cases}$$

Examples

- **Example 4.18** A Markov chain consists of the ~~four~~ five states 0, 1, 2, 3, 4 and have transition probability matrix

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0. & 0.5 & 0.5 & 0 \\ 0 & 0. & 0.5 & 0.5 & 0 \\ 0.25 & 0.25 & 0 & 0 & 0.5 \end{pmatrix}.$$

  What are its classes? Are they recurrent or transient?

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0. & 0.5 & 0.5 & 0 \\ 0 & 0. & 0.5 & 0.5 & 0 \\ 0.25 & 0.25 & 0 & 0 & 0.5 \end{pmatrix} \end{array}.$$

$0 \to 1$

$1 \to 0$

$2 \to 3$

$3 \to 2$

$4 \to 0, \; 4 \to 1, \; 4 \to 4$

The chain has 3 classes: $\{0,1\}$, $\{2,3\}$, $\{4\}$.

recurrent, recurrent, transient.

$$p_{44}^n = P(X_n = 4 \mid X_0 = 4) = P(X_n = X_{n-1} = \cdots = X_1 = 4 \mid X_0 = 4)$$

$$= \left(\frac{1}{2}\right)^n$$

Examples

- **Example 4.19** Are the states of a random walk recurrent or transient?

$d = 1, 2$, the random walk is recurrent

$d \geq 3$, " " " " transient

Let $X_1, X_2, X_3, \cdots$ be iid r.v.'s taking 1 and -1 with probab p and $q = 1-p$. Then $S_n = X_1 + X_2 + \cdots + X_n$ is the position of the random walk at time n after starting at 0.

Thus $P(S_n = 0) = 0$ if n is odd and

$$P(S_{2n} = 0) = P(n \text{ of } X_i = 1 \text{ and the remaing } = -1)$$

$$= \binom{2n}{n} p^n q^n$$

By the stirling formula, $n! = \sqrt{2\pi n}\, n^n e^{-n}$, we get

$$\binom{2n}{n} = \frac{(2n)!}{n!\,n!} \sim \frac{2^{2n}}{\sqrt{n\pi}}.$$

$a_n \sim b_n$: $\frac{a_n}{b_n} \to 1$

Thus,

$$P(S_{2n} = 0) \sim \frac{(4pq)^n}{\sqrt{n\pi}}$$

Obviously, $\sum_{n=1}^{\infty} P(S_n = 0) = \frac{1}{\sqrt{\pi}} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} (4pq)^n$

$$\begin{cases} = \infty & \text{iff } p = \frac{1}{2} \cdot \text{ Recurrent} \\ < \infty & \text{iff } p \neq \frac{1}{2} \cdot \text{ Transient} \end{cases}$$

From here to next page, no need to read it!

Similarly, for 2-dim case, $X_i = (X_i^1, X_i^2)$, symmetric random walk

$$P(S_{2n} = 0) = \sum_{k=0}^{n} P\Big(k \text{ of } X_i^1 = 1,\ k \text{ of } X_i^1 = -1,\ n-k \text{ of } X_i^2 = 1,\ n-k \text{ of } X_i^2 = -1\Big)$$

$$= \sum_{k=0}^{n} \binom{2n}{k, k, n-k, n-k} \frac{1}{4^{2n}}$$

$$= \frac{1}{4^{2n}} \sum_{k=0}^{n} \frac{(2n)!}{k!\,k!\,(n-k)!\,(n-k)!}$$

$$\therefore \quad P(S_{2n}=0) = \frac{1}{4^{2n}} \frac{(2n)!}{(n!)^2} \sum_{k=0}^{n} \binom{n}{k}^2$$

$$= \frac{1}{4^{2n}} \frac{(2n)!}{(n!)^2} \binom{2n}{n}$$

$$= \frac{1}{4^{2n}} \binom{2n}{n}^2 \sim \frac{1}{n\pi}$$

$$\therefore \quad \sum_{n=1}^{\infty} P(S_n = 0) = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

For 3-dim case, $X_i = (X_i^1, X_i^2, X_i^3)$.

$$P(S_{2n}=0) = \sum_{0 \le k+l \le n} P\left(\begin{array}{c} k \text{ of } X_i^1 = 1,\ k \text{ of } X_i^1 = -1, \\ l \text{ of } X_i^2 = 1,\ l \text{ of } X_i^2 = -1, \\ n-k-l \text{ of } X_i^3 = 1,\ n-k-l \text{ of } X_i^3 = -1 \end{array} \right)$$

$$= \sum_{0 \le k+l \le n} \binom{2n}{k, k, l, l, n-k-l, n-k-l} \frac{1}{6^{2n}}$$

$$= \sum_{0 \le k+l \le n} \frac{(2n)!}{(k!)^2 (l!)^2 (n-k-l)!^2} \frac{1}{6^{2n}}$$

$$= \frac{1}{6^{2n}} \cdot \frac{(2n)!}{(n!)^2} \sum_{0 \le k+l \le n} \left[\frac{n!}{k!\, l!\, (n-k-l)!} \right]^2$$

$$\le \underbrace{\frac{1}{6^{2n}} \cdot \frac{2^{2n}}{\sqrt{n\pi}}}_{} \quad \underbrace{\max_{0 \le k+l \le n} \frac{n!}{k!\, l!\, (n-k-l)!}}_{\le C \frac{3^n}{n}} \cdot \underbrace{(1+1+1)^n}_{3^n}$$

$$\le C \cdot \frac{1}{n^{3/2}}.$$

Thus, at $d=3$, $\sum_{n=1}^{\infty} P(S_{2n}=0) < \infty$. Transient!

Examples

- What about the Markov chain in the gambling model?

Transition matrix, states: $0, 1, 2, \cdots, N$

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ N \end{array} \begin{array}{c} \begin{array}{ccccccc} 0 & 1 & 2 & 3 & & & N \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & & 0 \\ p & 0 & q & 0 & \cdots & & 0 \\ 0 & p & 0 & q & \cdots & & 0 \\ 0 & 0 & p & 0 & \cdots & & 0 \\ \vdots & \vdots & \vdots & \vdots & & & \vdots \\ 0 & 0 & 0 & 0 & \cdots & & 1 \end{pmatrix} \end{array}$$

$p = P(\text{lose } \$1)$

$q = P(\text{win } \$1)$

3 classes: $\{0\}$, $\{1, 2, \cdots, N-1\}$, $\{1\}$

$\{0\}$: recurrent

$\{1, 2, \cdots, N-1\}$: transient

$\{1\}$: recurrent

Long-run Proportions

- For any pair of states $i \neq j$, define

$$f_{i,j} = P(X_n = j \text{ for some } n > 0 | X_0 = i).$$

- **Proposition 4.3** If i is recurrent and i communicates with j, then $f_{i,j} = 1$.
- For a recurrent state j, we define

$$N_j = \min\{n > 0 : X_n = j, \text{given } X_0 = j\},$$

and

$$m_j = E[N_j | X_0 = j].$$

- We say the recurrent state j is positive recurrent if $m_j < \infty$ and say that it is null recurrent if $m_j = \infty$.

Long-run Proportion

- Let π_j denote the long-run proportion of time that the Markov chain is in state j.
- **Proposition 4.4** If a Markov chain is irreducible and recurrent, then for any initial state

$$\pi_j = 1/m_j.$$

$\pi_j = 0$ if null recurrent

- **Example:** Consider a two-state Markov chain with transition probability

$$P = \begin{Vmatrix} 0 & 1 \\ 1 & 0 \end{Vmatrix}.$$

Let the Markov chain run m times starting from a certain state. Let j_m be the number of times the chain visits j. Then

$$\pi_j = \lim_{n \to \infty} \frac{j_m}{m}.$$

Proof of Prop 4.4. Let the chain start at i.

T_1 = 1st time the chain visits j.

T_2 = 2nd - - - -

$\vdots$

T_n = nth - - - - - -

By the Markov property $T_2, T_3, \dots$ are iid and $E T_2 = m_j$. Thus

$$\pi_j = \lim_{n\to\infty} \frac{n}{T_1 + T_2 + \cdots + T_n}$$

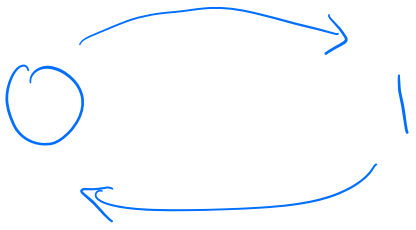
$$= \lim_{n\to\infty} \frac{1}{\frac{T_1}{n} + \frac{T_2 + \cdots + T_n}{n}}$$

$\frac{T_1}{n} \to 0$

$\frac{T_2 + \cdots + T_n}{n} \to E T_2 = m_j$ law of large numbers

$$= \frac{1}{m_j}$$

Example. Markov chain



$$P(0,1) = P(1,0) = 1$$

Transition matrix: $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Easily, $\pi_0 = \pi_1 = \frac{1}{2}$.

m_0 = the 1st time arriving 0 after starting at state 0

$= 2$.

Long-run Proportion

- **Theorem 4.1** Consider an irreducible Markov chain. If the chain is positive recurrent, the long-run proportions are the unique solution of the equations

$$\pi_j = \sum_i \pi_i P_{i,j}, j \geq 1$$

$$\sum_j \pi_j = 1.$$

Moreover, if there is no solution of the above linear equations, the Markov chain is either transient or null recurrent, and all $\pi_j = 0$.

Let $\pi = (\pi_0, \pi_1, \pi_2, \cdots)$. Then

$$\begin{cases} \pi = \pi P \\ \pi \mathbf{1} = 1 \end{cases}$$

P is transition matrix, i.e., $P = (P_{ij})$

where $\mathbf{1} = (1, 1, 1, \cdots)^T$

In particular, $\pi = \pi P = \pi P^2 = \pi P^3 = \cdots = \pi P^n = \cdots$

Long-run Proportion

- **Proposition 4.5** If i is positive recurrent and communicates with j, then j is positive recurrent.

As a consequence,

Positive recurrence is a class property;

Null " " " also a " " "

- **Corollary:** An irreducible finite state Markov chain must be positive recurrent.

Proof. A finite state irreducible MC is always recurrent.

If one state is null recurrent, then all states are null recurrent. Thus, $\pi_j = 0$, $\forall j$. Contradicts $\sum_j \pi_j = 1$.

- **Proposition 4.5** If i is positive recurrent and communicates with j, then j is positive recurrent.

Lemma 1. Let X_0, X_n be r.v.'s s.t. $X_n \to X_0$ in probab, i.e., $\lim_n P(|X_n - X_0| \geq \varepsilon) = 0$ for any $\varepsilon > 0$. Assume $|X_n| \leq a$ for all $n \geq 0$ where a is a const not depending on n. Then $\lim_{n\to\infty} EX_n = EX_0$. The conclusion also holds if $X_n \to X_0$ a.s.

Proof. Notice $|EX_n - EX_0| \leq E|X_n - X_0|$. Set $A = \{|X_n - X_0| \geq \varepsilon\}$. Then

$$E|X_n - X_0| = E(|X_n - X_0| I_A) + E(|X_n - X_0| I_{A^c})$$
$$\leq 2a \cdot P(|X_n - X_0| \geq \varepsilon) + \varepsilon$$

for any $\varepsilon > 0$. Let $n \to \infty$, then let $\varepsilon \to 0$ we get $E|X_n - X_0| \to 0$. Since $X_n \to X_0$ a.s. implies $X_n \to X_0$ in probab, the 2nd conclusion holds.

Proof of Prop 4.5. By assumption, $\pi_i > 0$ and $p_{ij}^m > 0$ for some $m \geq 1$. By def.

$$\pi_j = \lim_{T\to\infty} \frac{1}{T} \sum_{n=1}^{T} I(X_n = j)$$

Notice $0 \leq \frac{1}{T}\sum_{n=1}^{T} I(X_n = j) \leq 1$. By Lemma 1,

$$\pi_j = \lim_T \frac{1}{T} \sum_{n=1}^{T} P(X_n = j) \geq \lim_T \frac{1}{T} \sum_{n=m}^{T} P(X_n = j \mid X_{n-m} = i) P(X_{n-m} = i)$$
$$= \lim_T \frac{1}{T} \sum_{n=m}^{T} p_{ij}^m \cdot P(X_{n-m} = i) = p_{ij}^m \cdot \pi_i > 0.$$

Since $\lim_T \frac{1}{T} \sum_{n=m}^{T} P(X_{n-m} = i) \overset{k=n-m}{=} \lim_{T\to\infty} \frac{T-m}{T} \cdot \frac{1}{T-m} \sum_{k=0}^{T-m} P(X_k = i) = \pi_i$.

Examples

- **Example** The symmetric random walk is null recurrent.

- **Example 4.23** Consider Example 4.3 in which the mood of an individual is considered as a three-state Markov chain having a transition probability matrix

$$P = \begin{Vmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{Vmatrix}$$

  In the long run, what proportion of time is the process in each of the three states?

Lemma

Example 2.7. Consider a gambler who in each game is equally likely to either win or lose 1, independent of the results from earlier games. Starting with a fortune of i, find m_i, the mean number of games until the gambler's fortune is either 0 or n, where $0 \le i \le n$. $m_i = i(n-i)$.

Solution. Set

N = # of games until the gambler's fortune is 0 or n.

S_i = the event the gambler starts with i dollars.

Then $m_i = E(N|S_i)$. obviously, $m_0 = m_n = 0$.

Let $W = \{1\text{st game is win}\}$.

$L = \{ - - - \cdots \text{lose}\}$.

Then

$$m_i = E(N|S_iW)P(W|S_i) + P(N|S_iL)P(L|S_i)$$

$$= \frac{1}{2}(1 + m_{i+1}) + \frac{1}{2}(1 + m_{i-1})$$

Thus, $m_{i+1} = 2m_i - m_{i-1} - 2$, $1 \le i \le n-1$.

With initial condition $m_0 = m_n = 0$, we get

$$m_i = i(n-i), \quad 0 \le i \le n.$$

Examples

- **Example** The symmetric random walk is null recurrent.

Solution: For any i, j, Let

$X_{i,j}$ = # of transitions, starting at i, 1st time to visit j.

Let $m_{i,j} = E X_{i,j}$. In the above we showed the mean time to visit 0 or n, starting from 1, is $n-1$.

Then $m_{1,0} \geq n-1$, $\forall n$. Thus $m_{1,0} = \infty$.

Also,

$$m_{0,0} = E(X_{0,0} \mid 1\text{st move is left}) \cdot P(1\text{st move is left} \mid X_0 = 0)$$
$$+ \; ---\text{right}) \cdot \; ---\text{right} \cdots$$
$$= \frac{1}{2}(1 + m_{-1,0}) + \frac{1}{2}(1 + m_{1,0})$$

$\therefore \quad m_{0,0} = \infty.$

$\therefore \quad \pi_0 = \frac{1}{m_{0,0}} = 0.$ Null recurrent!

▸ **Example 4.23** Consider Example 4.3 in which the mood of an individual is considered as a three-state Markov chain having a transition probability matrix

$$P = \begin{Vmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{Vmatrix}$$

In the long run, what proportion of time is the process in each of the three states?

Calculate π_0, π_1, π_2

Solution. Now

$$(\pi_0, \pi_1, \pi_2) = (\pi_0, \pi_1, \pi_2) \begin{pmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

or equivalently,

$$\begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.4 & 0.3 \\ 0.1 & 0.3 & 0.5 \end{pmatrix} \begin{pmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{pmatrix} = \begin{pmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{pmatrix}.$$

Use $\pi_0 + \pi_1 + \pi_2 = 1$ to get

$$\pi_0 = \frac{21}{62}, \quad \pi_1 = \frac{23}{62}, \quad \pi_2 = \frac{18}{62}.$$